



Student Number: _____

St Catherine's School Waverley

August 2011

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question
- Complete each question in a separate booklet
- Attempt Questions 1 – 8
- All questions are of equal value
- Total Marks – 120

Total marks -120
Attempt Questions 1-8
All questions are of equal value

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 1 (15 marks) Use the Question 1 Writing Booklet. Marks

(a) Find $\int \frac{dx}{x^2 - 2x + 5}$ 2

(b) Use integration by parts to evaluate $\int_0^1 \tan^{-1} x \, dx$ 3

(c) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x + \cos x}$ 3

(d) Find the values of a , b and c such that; 2

$$\frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} = \frac{a}{2x - 1} + \frac{bx + c}{x^2 + 4}$$

(ii) Hence evaluate $\int \frac{x^2 - x - 21}{(2x - 1)(x^2 + 4)} dx$ 2

(e) Use the substitution $x = \frac{\pi}{2} - u$ to show that 3

$$\int_0^{\frac{\pi}{2}} \frac{\cos x - \sin x}{1 + \sin 2x} dx = 0$$

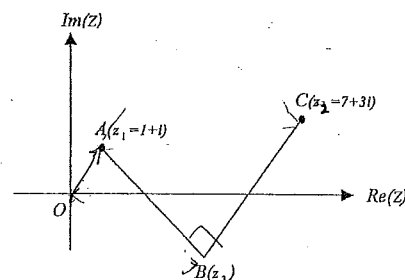
Question 2 (15 marks) Use the Question 2 Writing Booklet.

Marks

- (a) (i) Express $z = \sqrt{3} + i$ in modulus-argument form. 1
- (ii) Hence show that $z^7 + 64z = 0$ 2
- (b) On an argand diagram the point P representing the complex number z moves such that $|z - (1 + i)| = 1$
- (i) Sketch the locus of P 1
- (ii) Find the greatest value of $|z|$ 2
- (iii) Shade the region common to $|z - (1 + i)| \leq 1$ and $0 < \arg(z - 1) < \frac{\pi}{4}$ 2
- (iv) Find the area of the region in part (iii) above 2
- (c) If w is one of the complex roots of $z^3 = 1$
- (i) Show that w^2 is also a root. 1
- (ii) Show that $1 + w + w^2 = 0$ 1
- (iii) Evaluate $(1 - w)(1 - w^2)(1 - w^4)(1 - w^8)$ 3

Question 3 (15 marks) Use the Question 3 Writing Booklet.

Marks

- (a) Given that $(x + i)$ is a factor of $P(x) = x^4 + 3x^3 + 6x^2 + 3x + 5$ factorise $P(x)$ over the complex field. 4
- (b) Given that the equation $x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$ has a root of multiplicity 3, find all the roots of this equation. 3
- (c) If α, β, γ are the roots of $x^3 - 3x^2 + 2x - 1 = 0$ find the equation whose roots are
- (i) $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$ 2
- (ii) $\alpha^2, \beta^2, \gamma^2$ 3
- (d)
- 
- Diagram not to scale
- The points A and C represent the complex numbers $z_1 = 1 + i$ and $z_2 = 7 + 3i$.
- Find the complex number z_3 , represented by the point B such that $\triangle ABC$ is isosceles and right angled at B . 3

Question 4 (15 marks) Use the Question 4 Writing Booklet.

Marks

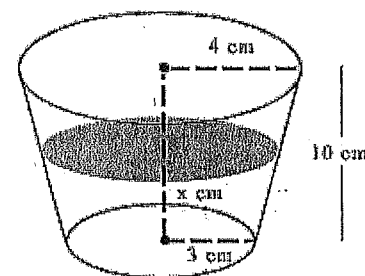
The equation of an ellipse E is given by $\frac{x^2}{9} + \frac{y^2}{5} = 1$

- (i) Find the eccentricity of E 1
- (ii) Write down
- (a) The coordinates of the foci 1
- (b) The equations of the directrices 1
- (c) The equation of the major auxiliary circle A 1
- (iii) Draw a neat sketch of E and A showing clearly the features in part (b) above (at least one third of a page) 2
- (iv) A line parallel to the y -axis meets the x -axis at N and the curves E and A at P and Q respectively. If N has coordinates $(3\cos\theta, 0)$ and given that P and Q are in the first quadrant, show that the coordinates of P are $(3\cos\theta, \sqrt{5}\sin\theta)$ and the coordinates of Q are $(3\cos\theta, 3\sin\theta)$. 2
- (v) Show that the equations of the tangents at P and Q are $\sqrt{5}\cos\theta x + 3\sin\theta y = 3\sqrt{5}$ and $x\cos\theta + y\sin\theta = 3$ respectively. 4
- (vi) Show that the point of intersection R of these tangents lies on the major axis of E produced. 1
- (vii) Prove that $ON \cdot OR$ is independent of the position of P and Q on the curves. 2

Question 5 (15 marks) Use the Question 5 Writing Booklet.

Marks

- (a) Prove that the area bounded by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab square units 3
- (b) Use the method of cylindrical shells, find the volume of the solid formed when the area in (a) is rotated through one complete revolution about the line $y = b$ 3
- (c) A drinking glass is in the shape of a truncated cone, in which the internal diameter of the top and the bottom are 8 cm and 6 cm respectively. 3



- (i) If the internal height of the glass, MN , is 10 cm, show that the area of the cross-section at a height of x cm above the base is $\pi\left(3 + \frac{x}{10}\right)^2 \text{ cm}^2$ 3
- (ii) Hence find by integration, the volume of the glass. 2
- (c) The points $P(a\sec\theta, b\tan\theta)$ and $Q(a\sec\phi, b\tan\phi)$ lie on the same branch of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ and PQ is a focal chord, passing through $S(ae, 0)$. 4

Use the gradients of PS and QS to show that $e = \frac{\sin\theta - \sin\phi}{\sin(\theta - \phi)}$

Question 6 (15 marks) Use the Question 6 Writing Booklet.

Marks

- (a) (i) Show that the equation of the normal to the hyperbola $xy = c^2$ at

2

$$P\left(cp, \frac{c}{p}\right) \text{ is } p^3x - py = c(p^4 - 1)$$

- (ii) The normal at $P\left(cp, \frac{c}{p}\right)$ meets the x -axis at Q . Find the coordinates of Q .

1

- (iii) Find the coordinates of the mid point, R , of PQ .

1

- (iv) Hence find the equation of the locus of R .

2

- (b) Use the compound angle formula for $\cos(x+y)$ and $\cos(x-y)$ to prove

2

$$\text{the result } \cos S - \cos T = -2 \sin\left(\frac{S+T}{2}\right) \sin\left(\frac{S-T}{2}\right)$$

- (c) If I_n is defined such that $I_n = \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2nx}{\sin 2x} dx$ for $n = 0, 1, 2, 3, \dots$

$$\text{Show that } I_1 = \frac{1}{2} \ln 2$$

2

- Using the result proven in part (b) above, show that for $r \geq 1$;

3

$$I_{2r+1} - I_{2r-1} = \frac{1 - (-1)^r}{2r}$$

- (iii) Hence evaluate I_5

2

Question 7 (15 marks) Use the Question 7 Writing Booklet.

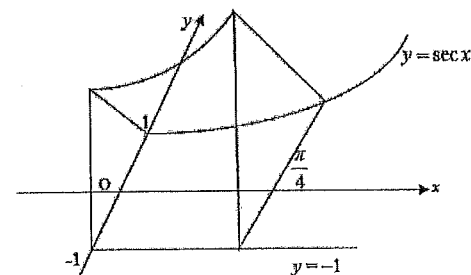
Marks

- (a) A string 50 cm in length can just sustain a weight of mass 20 kg without breaking. A mass of 4 kg is attached to one end of the string and revolves *uniformly* on a smooth horizontal table. The other end is fixed to a point on the table.

3

Find the greatest number of complete revolutions the mass can make in a minute without breaking the string. [Use acceleration due to gravity as 10 ms^{-2}]

- (b) The base of a solid is the region in the xy plane enclosed by the curve $y = \sec x$, $y = -1$, $x = 0$ and $x = \frac{\pi}{4}$. Each cross-section perpendicular to the x -axis is an equilateral triangle.



- (i) Show that the area of the triangular cross-section x units from the origin is given by

1

$$A = \frac{\sqrt{3}}{4} (\sec x + 1)^2$$

- (ii) Show by differentiation that $\int \sec x \, dx = \ln(\sec x + \tan x) + c$

1

- (iii) Hence, show that the volume of the solid is given by;

3

$$\frac{\sqrt{3}}{4} \left[1 + 2 \ln(\sqrt{2} + 1) + \frac{\pi}{4} \right] \text{ units}^3$$

- (c) If $U_1 = 1$, $U_2 = 5$ and $U_n = 5U_{n-1} - 6U_{n-2}$ for $n \geq 3$, prove by mathematical induction that $U_n = 3^n - 2^n$ for $n \geq 1$

4

- (d) The complex number z moves so that the sum of its distances from 3 ($|z - 3|$) and -3 ($|z + 3|$) is 10 units. Find the Cartesian equation of the ellipse described by the locus of z

3

Question 8 (15 marks) Use the Question 8 Writing Booklet.

Marks

- (a) (i) Given that a , b , and c are three non negative numbers, show that the arithmetic mean $\geq$ geometric mean. 3

i.e. show that $\frac{a+b+c}{3} \geq \sqrt[3]{abc}$

(You may assume that $a^2 + b^2 \geq 2ab$)

- (ii) Given that $a + \frac{1}{a} \geq 2$ show that $(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) \geq 9$ 2

- (b) A body of unit mass falls under gravity through a resisting medium. The body falls from rest from a height of 50 metres above the ground. The resistance to its motion is $\frac{1}{100}v^2$ where v metres per second is the speed of the body when it has fallen a distance of x metres. The acceleration due to gravity is $g \text{ ms}^{-2}$

- (i) Show that the equation of motion of the body is : 2

$$\ddot{x} = g - \frac{1}{100}v^2$$

- (ii) Show that the terminal velocity V of the body is given by 1

$$V = \sqrt{100g}$$

- (iii) Hence show that $v^2 = V^2(1 - e^{-\frac{x}{50}})$ 3

- (iv) Find the distance fallen in metres until the body reaches a velocity equal to 50% of the terminal velocity. 2

- (v) Find the velocity reached as a percentage of terminal velocity when the body hits the ground. 2

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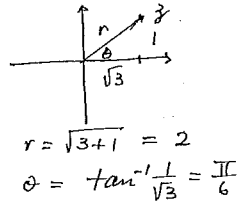
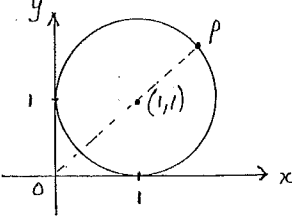
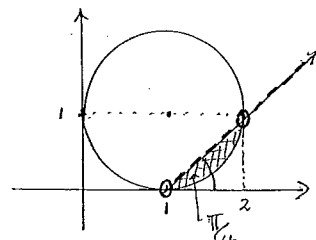
Q	Solutions	Marks	Comments
Q1a)	$\int \frac{dx}{x^2 - 2x + 5}$ $= \int \frac{dx}{(x^2 - 2x + 1) + 4}$ $= \int \frac{dx}{(x-1)^2 + 2^2}$ $= \frac{1}{2} \tan^{-1}\left(\frac{x-1}{2}\right) + C$	1	
b)	$\int_0^1 \tan^{-1}x \, dx \quad u = \tan^{-1}x \quad v = x$ $u' = \frac{1}{1+x^2} \quad v' = 1$ $= \int_0^1 \tan^{-1}x \, dx = [x \tan^{-1}x]_0^1 - \int_0^1 \frac{x}{1+x^2} \, dx$ $= \frac{\pi}{4} - \left[\frac{1}{2} \ln(1+x^2) \right]_0^1$ $= \frac{\pi}{4} - \frac{1}{2} \ln 2$	1	
c)	$\int_0^{\frac{\pi}{2}} \frac{dx}{1 + \sin x + \cos x} \quad t = \tan \frac{x}{2} \quad \sin x = \frac{2t}{1+t^2}$ $dx = \frac{2dt}{1+t^2} \quad \cos x = \frac{1-t^2}{1+t^2}$ $= \int_0^1 \frac{2dt}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}$ $x=0 \quad t=0$ $x=\frac{\pi}{2} \quad t=1$ $= \int_0^1 \frac{2dt}{1+t^2+2t+1-t^2}$ $= 2 \int_0^1 \frac{dt}{2t+2}$ $= \int_0^1 \frac{dt}{t+1}$ $= [\ln(t+1)]_0^1$ $= \ln 2$	1	

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Q	Solutions	Marks	Comments
2a)	$(i) \quad z = \sqrt{3} + i$  $\therefore z = 2\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)$ $= 2 \operatorname{cis} \frac{\pi}{6}$	1	
(ii)	$z^7 = \left(2 \operatorname{cis} \frac{\pi}{6}\right)^7 = 2^7 \operatorname{cis} \frac{7\pi}{6}$ $64z = 128 \operatorname{cis} \frac{\pi}{6}$ $\therefore z^7 + 64z = 128 \operatorname{cis} \frac{7\pi}{6} + 128 \operatorname{cis} \frac{\pi}{6}$ $= 128 \left[\cos \frac{7\pi}{6} + i \sin \frac{7\pi}{6} + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right]$ $= 128 \left[-\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6} \right]$ $= 0$	1	
b)	$(i) \quad z - (1+i) = 1$ 	2	
(ii)	$ z $ is maximum when z is at P $\therefore$ Now $OP = \sqrt{2} + 1$ $\therefore$ max value of $ z = \sqrt{2} + 1$	1	
(iii)		2	
(iv)	$\text{Area shaded region} = \text{area of segment} = \frac{1}{2} \left(\frac{\pi}{2} - \sin \frac{\pi}{2} \right)$ $= \left(\frac{\pi}{4} - \frac{1}{2} \right) u^2$	1	

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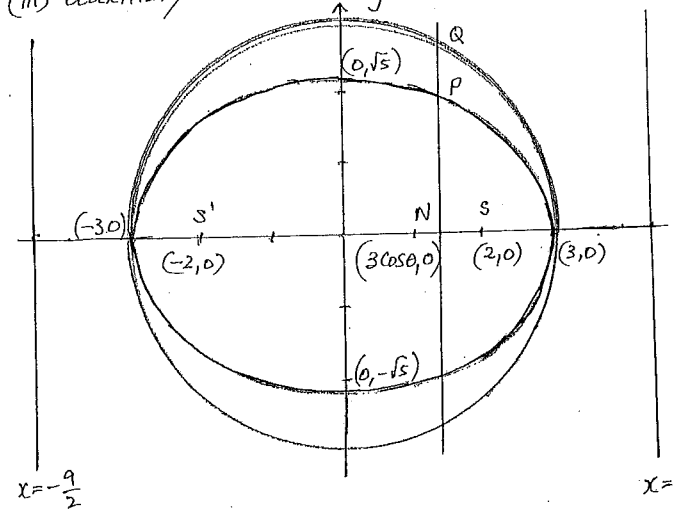
Q	Solutions	Marks	Comments
3a)	If $(x+i)$ is a factor of $P(x) = x^4 + 3x^3 + 6x^2 + 3x + 5$ then $(x-i)$ is also a factor of $P(x)$ ($P(x)$ real coefficients) $\therefore (x+i)(x-i) = (x^2+1)$ is also a factor Now $x^4 + 3x^3 + 6x^2 + 3x + 5 = (x^2+1)(x^2+3x+5)$ $\therefore$ the zeros of $P(x)$ are $i, -i, \frac{-3 \pm \sqrt{11}i}{2}$ $\therefore P(x) = (x-i)(x+i)(x + \frac{-3 - \sqrt{11}i}{2})(x + \frac{-3 + \sqrt{11}i}{2})$	1 1 1 1	
b)	$P(x) = x^4 - 5x^3 - 9x^2 + 81x - 108 = 0$ $P'(x) = 4x^3 - 15x^2 - 18x + 81$ $P''(x) = 12x^2 - 30x - 18 = 0$ for possible roots of multiplicity 3. $\therefore 2x^2 - 5x - 3 = 0$ $(2x+1)(x-3) = 0$ $\therefore x = -\frac{1}{2}, 3$ Now $P'(3) = 108 - 135 - 54 + 81 = 0$ and $P(3) = 81 - 135 - 81 + 243 - 108 = 0$ $\therefore x = 3$ is a root of multiplicity 3. $\therefore (x-3)^3$ is a factor of $P(x)$ also, roots are $3, 3, 3, \alpha$ Now product of roots $= 27\alpha = -108$ $\therefore \alpha = -4$ $\therefore$ the roots of $P(x) = 0$ are $x = 3, 3, 3, -4$	1 1 1 1 1 1 1 1	
c)	$x^3 - 3x^2 + 2x - 1 = 0$ α, β, γ are roots (i) let $y = \frac{1}{x} \therefore x = \frac{1}{y}$ $\therefore (\frac{1}{y})^3 - 3(\frac{1}{y})^2 + 2(\frac{1}{y}) - 1 = 0$ $\frac{1}{y^3} - \frac{3}{y^2} + \frac{2}{y} - 1 = 0$ $\therefore 1 - 3y + 2y^2 - y^3 = 0$ which equates to the polynomial in x of $x^3 - 2x^2 + 3x - 1 = 0$	1 1 1 1 1 1 1 1	

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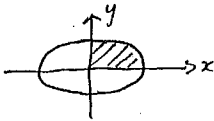
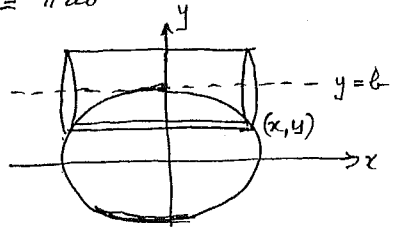
Q	Solutions	Marks	Comments
Q4a)	$\frac{x^2}{9} + \frac{y^2}{5} = 1$ for ellipse $b^2 = a^2(1-e^2)$ $\therefore 5 = 9(1-e^2)$ $\therefore e^2 = \frac{4}{9}$ $e = \frac{2}{3}$	1 1 1 1	
b)	(i) foci $(\pm ae, 0)$ i.e. $(\pm 2, 0)$ (ii) directrices: $x = \pm \frac{a}{e}$ i.e. $x = \pm \frac{9}{2}$ (iii) auxiliary circle: $x^2 + y^2 = 9$	1 1 1	
c)		2	
d)	Coordinates of Q $(3 \cos \theta, 3 \sin \theta)$ Coordinates of P $(3 \cos \theta, \sqrt{5} \sin \theta)$	1 1	
e)	at P $x = 3 \cos \theta$ $y = \sqrt{5} \sin \theta$ $\frac{dx}{d\theta} = -3 \sin \theta$ $\frac{dy}{d\theta} = \sqrt{5} \cos \theta$ Now $\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$ $= \frac{-\sqrt{5} \cos \theta}{3 \sin \theta}$	1 1 1	

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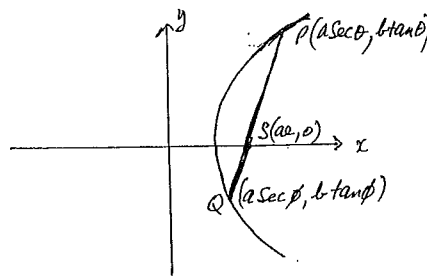
Q	Solutions	Marks	Comments
5a)	$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ $y^2 = b^2(1 - \frac{x^2}{a^2})$ $y^2 = \frac{b^2}{a^2}(a^2 - x^2)$ $y = \pm \frac{b}{a} \sqrt{a^2 - x^2}$ Shaded area is given by $A = \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$ $= \frac{b}{a} \int_0^a \sqrt{a^2 - x^2} dx$ now $\int_0^a \sqrt{a^2 - x^2} dx$ is a quadrant of a circle radius a $= \frac{1}{4} \pi a^2$ $\therefore A = \frac{b}{a} \times \frac{1}{4} \pi a^2$ $= \frac{1}{4} \pi ab$ $\therefore \text{area of ellipse} = \pi ab$ 	1	could also be done by substitution let $u = a \sin \theta$
b)	radius of shell is $(b-y)$ height of shell is $2x$  $\therefore \delta V = 2\pi (b-y) \cdot 2x$ $\therefore V = 4\pi \int_{-b}^b (b-y) \cdot \frac{a}{b} \sqrt{b^2 - y^2} dy$ $= \frac{4\pi a}{b} \int_{-b}^b (b\sqrt{b^2 - y^2} - y\sqrt{b^2 - y^2}) dy$	1	

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Q	Solutions	Marks	Comments
Q5d)	gradient of PS: $\frac{b \tan \theta - 0}{a \sec \theta - ae}$ gradient of SQ: $\frac{b \tan \phi - 0}{a \sec \phi - ae}$ $\therefore \frac{b \tan \theta}{a \sec \theta - ae} = \frac{b \tan \phi}{a \sec \phi - ae}$ $ab \tan \theta \sec \phi - ab \tan \theta = ab \tan \phi \sec \theta - ab \tan \phi$ $\therefore ab \tan \phi - ab \tan \theta = ab \tan \phi \sec \theta - ab \tan \theta \sec \phi$ $\therefore e = \frac{ab(\tan \phi \sec \theta - \tan \theta \sec \phi)}{ab(\tan \phi - \tan \theta)}$ $= \frac{\frac{\sin \phi}{\cos \phi} \cdot \frac{1}{\cos \theta} - \frac{\sin \theta}{\cos \theta} \cdot \frac{1}{\cos \phi}}{\frac{\sin \phi}{\cos \phi} - \frac{\sin \theta}{\cos \theta}}$ $= \frac{\sin \phi \cos \theta - \sin \theta \cos \phi}{\sin \phi \cos \theta - \sin \theta \cos \phi}$ $= \frac{\sin \phi - \sin \theta}{\sin(\phi - \theta)}$ 	1	

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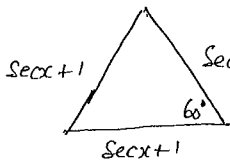
Q	Solutions	Marks	Comments
86b)	$\cos(x+y) = \cos x \cos y - \sin x \sin y$ — (1) $\cos(x-y) = \cos x \cos y + \sin x \sin y$ — (2) let $S = x+y$ and $T = x-y$ $S+T = 2x \therefore x = \frac{S+T}{2}$ $S-T = 2y \therefore y = \frac{S-T}{2}$ $\therefore (1)-(2) \cos S - \cos T = -2 \sin x \sin y$ i.e. $\cos S - \cos T = -2 \sin \frac{S+T}{2} \sin \frac{S-T}{2}$	1	
c)	$I_n = \int_0^{\pi/4} \frac{1 - \cos 2nx}{\sin 2x} dx \quad n = 0, 1, 2, 3, \dots$	1	
(i)	$I_1 = \int_0^{\pi/4} \frac{1 - \cos 2x}{\sin 2x} dx$ $= \int_0^{\pi/4} \frac{2 \sin^2 x}{2 \sin x \cos x} dx$ $= \int_0^{\pi/4} \frac{\sin x}{\cos x} dx$ $= -\int_0^{\pi/4} \ln \cos x dx$ $= -\ln \frac{1}{\sqrt{2}}$ $= \frac{1}{2} \ln 2$	1	
(ii)	$I_{2r+1} - I_{2r-1} = \int_0^{\pi/4} \frac{1 - \cos(4xr+2x)}{\sin 2x} dx - \int_0^{\pi/4} \frac{1 - \cos(4xr-2x)}{\sin 2x} dx$ $= \int_0^{\pi/4} \frac{\cos(4xr-2x) - \cos(4xr+2x)}{\sin 2x} dx$ $= \int_0^{\pi/4} \frac{-2 \sin 4xr \sin(-2x)}{\sin 2x} dx$ from (b) $= 2 \int_0^{\pi/4} \sin 4xr dx$ $= -\frac{1}{2r} [\cos 4xr]_0^{\pi/4} = -\frac{1}{2r} [\cos \pi r - 1]$ $= -\frac{1}{2r} [(-1)^r - 1] = \frac{1 - (-1)^r}{2r}$	1	

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Q	Solutions	Marks	Comments
7b)(i)	 $\text{Area } \Delta = \frac{1}{2} (\sec x + 1)^2 \sin 60^\circ$ $= \frac{1}{2} (\sec x + 1)^2 \frac{\sqrt{3}}{2}$ $= \frac{\sqrt{3}}{4} (\sec x + 1)^2$	1	
(ii)	$\frac{d}{dx} (\sec x + \tan x) = \frac{1}{\sec x + \tan x} (\sec x \tan x + \sec^2 x)$ $= \frac{\sec x (\tan x + \sec x)}{(\sec x + \tan x)}$ $= \sec x$	1	
(iii)	Volume of typical slice is given by $\delta V = \frac{\sqrt{3}}{4} (\sec x + 1)^2 \delta x$ $\therefore V = \frac{\sqrt{3}}{4} \int_0^{\pi/4} (\sec x + 1)^2 dx$ $= \frac{\sqrt{3}}{4} \int_0^{\pi/4} (\sec^2 x + 2 \sec x + 1) dx$ $= \frac{\sqrt{3}}{4} \left[\tan x + 2 \ln(\sec x + \tan x) + x \right]_0^{\pi/4}$ $= \frac{\sqrt{3}}{4} \left[1 + 2 \ln(\sqrt{2} + 1) + \frac{\pi}{4} - 0 \right]$ $\therefore V = \frac{\sqrt{3}}{4} \left[1 + 2 \ln(\sqrt{2} + 1) + \frac{\pi}{4} \right] \text{ units}^3$	1	

[illegible]

Q	Solutions	Marks	Comments
8b) (i)	$ma = mg - \frac{mv^2}{100}$ $\therefore a = g - \frac{v^2}{100}$ $\therefore \ddot{x} = g - \frac{v^2}{100}$ is the equation of motion	1	
(ii)	terminal velocity when $a = 0 \quad v = V$ $\therefore g - \frac{V^2}{100} = 0$ $\therefore V^2 = 100g$ $\therefore V = \sqrt{100g} \quad \text{also } g = \frac{V^2}{100}$	1	
(iii)	$\ddot{x} = v \frac{dv}{dx} = g - \frac{v^2}{100}$ from (ii) $g = \frac{V^2}{100}$ $\therefore v \frac{dv}{dx} = \frac{V^2}{100} - \frac{v^2}{100}$ $\therefore \frac{dv}{dx} = \frac{V^2 - v^2}{100v}$ $\therefore \frac{dx}{dv} = \frac{100v}{V^2 - v^2}$ $\therefore \int dx = \int \frac{100v}{V^2 - v^2} dv \quad (\text{integrating both sides})$ $\therefore x + C = -50 \ln(V^2 - v^2)$ when $v = 0 \quad x = 0 \Rightarrow C = -50 \ln V^2$ $\therefore x - 50 \ln V^2 = -50 \ln(V^2 - v^2)$ $x = 50 \ln V^2 - 50 \ln(V^2 - v^2)$ $\therefore \frac{x}{50} = \ln \left(\frac{V^2}{V^2 - v^2} \right)$ $\therefore \frac{V^2}{V^2 - v^2} = e^{\frac{x}{50}}$ $V^2 - v^2 = V^2 e^{-\frac{x}{50}}$ $v^2 = V^2 (1 - e^{-\frac{x}{50}})$	1	